
Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Exercise Series 1

Exercise 1. Consider the following model from optical and electrical communication:

$$y_t = a \cos(\omega t + \phi) + \epsilon_t, \quad t = 1, 2, \dots, n.$$

Here y_t 's are the observations at time $t = 1, 2, \dots, n$. This type of model is used to detect whether the successive observations contain a “signal” (some detectable pattern!) or not. If $a = 0$, then there is no signal present: the observations are completely random. The parameters of the model are a , called the “amplitude”, ω , called the “frequency” and ϕ , called the “phase”.

- (a) Is the given model a linear model? Recall that a model is linear if it is **linear in the parameters**.
- (b) Will your answer in part (a) change if ω is fixed (known)? After reparameterization (transformation of the parameters)?
- (c) Plot the function $a \cos(\omega t + \phi)$ for different values of a, ω and ϕ . Can you see why a is called the amplitude, ω is called the frequency and ϕ is called the phase?

Exercise 2. The previous exercise shows that sometimes a model that looks like nonlinear can be converted into a linear model by transforming the covariates and/or the parameters. Check whether the following models can be transformed into a linear model by transformation of the covariates and the parameters. In each case, clearly specify the new covariates and the new parameters.

(a) $y = \theta_0 + \frac{\theta_1}{x} + \frac{\theta_2}{x^2} + \epsilon$	(b) $y = \frac{\theta_0}{1 + \theta_1 x} + \epsilon$
(c) $y = \frac{\theta_0}{\theta_1 x} + \epsilon$	(d) $y = \frac{1}{\theta_0 + \theta_1 x + \epsilon}$
(e) $y = \theta_0 + \theta_1 x^{\theta_2} + \epsilon$	(f) $y = \theta_0 + \theta_1 x_1^{\theta_2} + \theta_3 x_2^{\theta_4} + \epsilon$
(g) $y = \theta_1 x_1^{\theta_2} (\cos x_2)^{\theta_3} \epsilon$	(h) $y = \theta_1 + x_1^{\theta_2} (2 + \cos x_2)^{\theta_3} (\epsilon^2 + 1)$

How would your conclusions change if you are also allowed to treat some (not all!) of the parameters as fixed?

Note: You can also transform the error term if needed. After all, the data consist of y and x 's, and the error term ϵ is only there on paper to signify the error that we may commit by using the particular model. But pay attention to the feasibility of the transformation! Is it possible to make the transformation you have made? Does it impose any restriction on the covariates/parameters/error?

Exercise 3 (Graded). Suppose that we have data on the salary of employees in a company and their educational qualification as shown in Table 1 below.

We want to see if (and how) the salary of an employee depends on her/his educational qualification. For this, we consider the following model for the salary (y):

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \epsilon, \quad \mathbb{E}(\epsilon) = 0. \tag{1}$$

Table 1: Salary and educational qualification of employees in a company (fictional).

Employee Id	Salary	Highest educational qualification
1201	50000	B. Sc. in Mathematics
1005	35000	Class XII
1001	80000	B. Com.
1135	90000	M. Sc.
1093	95000	MBA
1013	25000	Class VIII
1257	30000	Class XII
1178	20000	Class VIII

Here, x_1 is the “indicator” of completing high school education (i.e., $x_1 = 1$ if the employee has completed high school (class XII); otherwise $x_1 = 0$), x_2 is the indicator of completing college education (bachelor’s degree) and x_3 is the indicator of completing university education (master’s degree).

- Write the model (1) in the vector-matrix notation $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \epsilon$. Write down the response vector $\mathbf{y}$ and the data matrix $\mathbf{X}$ based on the observations in Table 1.
- What do the parameters of the model $\theta_0, \theta_1, \theta_2, \theta_3$ signify?
- Based on the model, how can you formulate the question: “does salary depend on the level of education”?

Now, consider a different model for the same data:

$$y = \gamma_1 x_1 + \gamma_2 x_2 + \gamma_3 x_3 + \epsilon. \quad (2)$$

- Write the new model (2) in vector-matrix notation $\mathbf{y} = \mathbf{X}'\boldsymbol{\gamma} + \epsilon$. What is the new data matrix $\mathbf{X}'$?
- What do the parameters of the new model $\gamma_1, \gamma_2, \gamma_3$ mean?
- Is there any connection between the two models (1) and (2)?
- Repeat the same task with the following models:

$$y = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \epsilon \quad (3)$$

$$y = \tau_0 + \tau_2 x_2 + \tau_3 x_3 + \epsilon \quad (4)$$

Note: The meaning of the error ϵ is different in the different models above.

Exercise 4. Suppose that our response variable y is binary, it takes two values 0 and 1 (or “success” and “failure”, or “happy” and “unhappy” etc.). In such a case, we are not interested in modeling y itself, but the probability of y being 1. Let p_i be the probability that the i -th observation is 1. We model this as a function of the covariates $x_1, \dots, x_p$ as:

$$p_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip})}}.$$

This model is known as the **logistic regression** model. Recall that for the binary variable y_i , $p_i = \mathbb{P}(y_i = 1) = \mathbb{E}(y_i)$. Thus, the model can also be written as

$$\mathbb{E}(y_i | x_{i1}, \dots, x_{id}) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_{i1} + \dots + \theta_d x_{id})}}.$$

This is NOT a linear model. But, if we take a transformation of $\mathbb{E}(y_i|x_{i1}, \dots, x_{id})$, then we get a linear model. Identify the transformation.

This exercise shows that sometimes it is also possible to transform the response variable (the conditional expectation of it) in order to have a linear model. Such models are called **generalized linear models**. We won't cover these models in our course.

You may now go back to Exercise 2 and check if some of those models can be converted into a linear model after transformation of the response. Pay attention to the feasibility of such transformations. For instance, what we have done in this exercise can only be done if the response is binary.

Exercise 5. This exercise is to make precise why we call a linear model “linear”. Recall that a model is linear if it is linear in the parameters. More precisely, the model is linear if $\mathbb{E}(y)$ as a function of the parameter $\boldsymbol{\beta}$ is a linear function in the sense that we have defined in the class:

$$f(c\boldsymbol{\beta}) = cf(\boldsymbol{\beta}) \quad \forall c \in \mathbb{R}, \boldsymbol{\beta} \in \mathbb{R}^d \quad \text{and} \quad f(\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2) = f(\boldsymbol{\theta}_1) + f(\boldsymbol{\theta}_2) \quad \forall \boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathbb{R}^d.$$

Consider the linear model

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i, \quad i = 1, \dots, n,$$

with $\mathbb{E}(\epsilon_i) = 0$, $i = 1, \dots, n$. Show that $\mathbb{E}(\mathbf{y})$ as a function of $\boldsymbol{\theta} = (\beta_0, \beta_1, \dots, \beta_p)^\top$ is a linear function. Hence the name.